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Project Title:

A Framework for Fake News Detection in Algerian Dialect Media

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***Note on Scope:** To ensure a coherent and focused narrative, a standalone System Design section was deemed outside the primary scope of this paper. Instead, critical system design elements were integrated directly into other relevant sections. This approach prioritizes high-impact findings and avoids repetitive content while maintaining the overall quality and technical depth of the work.*

NABA2: An Algerian Darija Corpus for Multi-Class Misinformation Detection

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Abstract

Misinformation on social media poses significant challenges in Arabic-speaking regions, particularly for low-resource dialects like Algerian Darija. This paper introduces a multi-class annotated corpus of 22,391 instances for misinformation detection in Algerian Darija, collected from social media platforms and news sources. Unlike existing binary datasets, our corpus distinguishes between five labels across several topic domains reflecting a spectrum of misinformation. We evaluate baseline models spanning classical machine learning, neural networks, and transformer architectures on the fake news detection task. Our results show that transformer models significantly outperform traditional approaches, with XLM-RoBERTa-Large achieving the best macro F1-score of 0.682. However, detecting Misleading content and Satire remains challenging across all architectures due to severe class imbalance and its inherently ambiguous nature. The corpus addresses a critical resource gap for Algerian Darija NLP and provides benchmarks for future misinformation detection research in dialectal Arabic.

Keywords: Algerian Darija, Corpus, Misinformation Detection, Fake News, Deep Learning, Transformer Models, Natural Language Processing

1 Introduction

The spread of misinformation on social media has become a pressing issue, with particularly damaging effects in Arabic-speaking regions (Azzeh et al., 2024). In the Arab world, online platforms are flooded daily with fabricated content—from celebrity death hoaxes to false claims about economic and diplomatic matters, creating a serious challenge for information integrity (Abdelouahab & Smaili, 2025).

The Arabic language itself makes this problem even more difficult to tackle. The wide variety of spoken dialects add layers of complexity that complicate efforts to detect and combat misinformation (Habash et al., 2023).

The Algerian dialect, known as Algerian Darija, is a particularly challenging case. It differs substantially from Modern Standard Arabic (MSA), blending vocabulary and grammar from French, local Berber tongues and most recently English code switching (Adouane et al., 2019), while often using informal, non-standardized spelling (Smaili et al., 2025). Unfortunately, there are few linguistic resources or annotated datasets available for Algerian Darija for this specific task with sufficient label granularity, which makes it extremely difficult to build effective natural language processing tools for misinformation detection in this low-resource dialect (Touileb & Barnes, 2021).

Beyond improving our understanding of how misinformation appears in dialectal Arabic, this study contributes practical resources that support the integrity of online communication in Arabic-speaking communities (Habash et al., 2023). In line with recent efforts to develop specialized linguistic resources for underrepresented varieties, such as the FASSILA corpus for Algerian dialect fake news detection and sentiment analysis, this work further enables more fine-grained analyses of deceptive content in dialectal Arabic discourse (Abdedaïem et al., 2024).

The remainder of this paper is organized as follows. Section 2 reviews related work on misinformation detection in Arabic and dialectal settings. Section 3 describes the construction of the proposed Algerian Darija corpus. Section 4 details the experimental setup, including model architectures, and evaluation metrics. Section 5 presents and analyzes the experimental results, providing comparisons across classical, neural, and transformer-based models, along with ablation and error analyses. Finally, Section 6 concludes the paper and outlines directions for future work.

2 Related Work

The rapid spread of misinformation in the digital age has prompted extensive research within the Natural Language Processing (NLP) community. Although fake news detection for English language content has reached a relatively mature stage, research on Arabic, especially its dialectal varieties, remains both limited and challenging. This section traces the progression of misinformation detection approaches, from early classical methods to recent transformer-based models, with particular emphasis on the unique challenges and emerging advances in Arabic and Algerian Dialect (AD) settings.

2.1 Arabic Misinformation Detection: From MSA to Dialects

Early efforts in Arabic fake news detection primarily targeted Modern Standard Arabic (MSA), utilizing high-resource datasets from official news agencies and verified media outlets. Azzeh et al. (2024) emphasize that while MSA provides a structured linguistic

framework, it fails to capture the informal, rapid-fire nature of misinformation often found on social media.

Recent studies have shifted focus toward more diverse data sources. For instance, the Arabic Fake News Dataset (AFND), introduced by (Khalil & et al., 2025), comprises over 600,000 articles from 134 news agencies, serving as a large-scale benchmark for the region. However, a recurring criticism of such corpora is their "formal" nature, which does not reflect the nuanced linguistic reality of the Maghreb region, where dialects like Algerian Darija predominate (Abdelouahab & Smaïli, 2025).

2.2 The FASSILA Corpus and Dialectal Resource Scarcity

The primary bottleneck in detecting misinformation in Algerian Darija has been the lack of annotated, high-quality datasets. (Touileb & Barnes, 2021) highlighted how the interplay between script (Arabic vs. Arabizi) and linguistic diversity complicates the development of multi-layered corpora.

A landmark contribution to this field is the **FASSILA** corpus (Abdedaïem et al., 2024). FASSILA addresses the critical shortage of resources by providing 10,087 annotated sentences in Algerian Dialect, covering seven distinct domains including politics, sports, and health. Unlike previous datasets, FASSILA is uniquely designed to handle both Fake News (FN) detection and Sentiment Analysis (SA). The methodology behind FASSILA involved:

- **Data Diversification:** Integration of social media comments from YouTube and Facebook with paraphrased MSA content using GPT-4 to ensure dialectal consistency.
- **Rigorous Labeling:** Annotations based on source credibility, logical consistency, and temporal relevance.
- **Quality Control:** Achieving high inter-annotator agreement (Cohen's Kappa), establishing it as a reliable ground truth for AD tasks.

Another recent effort is the **BOUTEF** corpus (Smaïli et al., 2025), which specifically targets the North African context (Algerian and Tunisian dialects). BOUTEF includes 3,600 instances of fake news posts and comments, specifically focusing on the "denial" category and the complex phenomenon of code-switching between Arabic, French, and English.

2.3 Evolution of Model Architectures

The transition from feature-engineered models to deep learning has significantly improved detection performance.

2.3.1 Classical and Neural Baselines

Traditional machine learning approaches, such as Support Vector Machines (SVM) and Random Forests (RF) using TF-IDF or word embeddings, provided the initial baselines for Arabic NLP. (Azzeh et al., 2024) demonstrated that while these models are computationally efficient, they struggle with the rich morphology and non-standardized

orthography of dialects. Hybrid architectures, such as CNN-BiLSTM models, were subsequently introduced to capture both local semantic features and long-term dependencies in news sequences, often achieving accuracies in the 80-90% range on standard datasets (Abdelouahab & Smaili, 2025).

2.3.2 Transformer-Based Models

The current state of the art is dominated by Pre-trained Language Models (PLMs). (Azzeh et al., 2024) show that fine-tuning transformer models like AraBERT, MARBERT, and QARiB significantly outperforms traditional methods. MARBERT, in particular, has shown superior performance on dialectal tasks because its pre-training data includes a massive corpus of social media text, which is linguistically closer to Darija than the formal news text used for AraBERT.

Recent advancements have explored **Ensemble Strategies** and **Consistency-Aware Merging**. (Khalil & et al., 2025) proposed a hybrid AraBERT+BiLSTM model that achieved an F1-score of 98% on the AFND dataset. Furthermore, research by (Abdedaïem et al., 2024) on the FASSILA corpus indicates that while transformers are powerful, their performance on AD is highly sensitive to the quality of the "dialectalization" of the input during the preprocessing phase.

2.4 Specific Challenges in Algerian Darija

The detection of fake news in Algeria is uniquely complicated by:

1. **Code-Switching:** The frequent intermixing of French and Arabic vocabulary, which requires models to have multilingual capabilities or specialized "code-switched" embeddings.
2. **Script Variation:** The use of both Arabic script and Arabizi (Latin script with numerals), necessitating models that are either script-agnostic or trained on multi-script data (Touileb & Barnes, 2021).
3. **Sarcasm and Context:** Misinformation in Darija often utilizes sarcasm or culture-specific metaphors that are difficult for standard sentiment or fake news classifiers to decode without deep contextual understanding (Abdedaïem et al., 2024).

In summary, while the field has moved toward high-performing transformer ensembles, the specific linguistic nuances of Algerian Darija necessitate specialized resources like FASSILA and BOUTEF. Our work builds upon these foundations by proposing a more granular analysis and a comparative evaluation of the latest transformer architectures specifically optimized for the Algerian social media ecosystem.

3 Corpus Building

Addressing the scarcity of multi-class annotated datasets for Algerian Darija identified in Section 2, we developed a corpus capturing the nuanced spectrum of misinformation types. Building such a dataset presents unique challenges: dialectal variability, limited resources, and misinformation's inherently ambiguous nature. Our systematic approach balances linguistic authenticity, annotation reliability, and practical usability to reflect the real information environment Algerian users encounter daily.

3.1 Data Collection Strategy

3.1.1 Linguistic Inclusivity

Algerian online content naturally mixes Darija, Modern Standard Arabic (MSA), French code-switching, and Arabizi (Latin-script Arabic). Rather than restricting collection to one variety, we adopted a spectrum-based approach capturing this diversity, ensuring models generalize to realistic scenarios.

3.1.2 Source Selection

Data collection occurred between October and November 2025 from Facebook, YouTube, Instagram, X (ex. Twitter), and online news platforms—sources characterized by rapid dissemination, limited moderation, and dialectal language where misinformation circulates most actively. Pages were manually chosen to represent both official and non-official media sources, including those known for disseminating misleading content and those recognized for anti-misinformation reporting. YouTube comments provided authentic conversational, reporting Darija.

3.1.3 Topic Diversity

Content spans healthcare, politics, general interest, crimes, and other domains . However, some Darija variations and certain labels remained underrepresented, necessitating LLM augmentation.

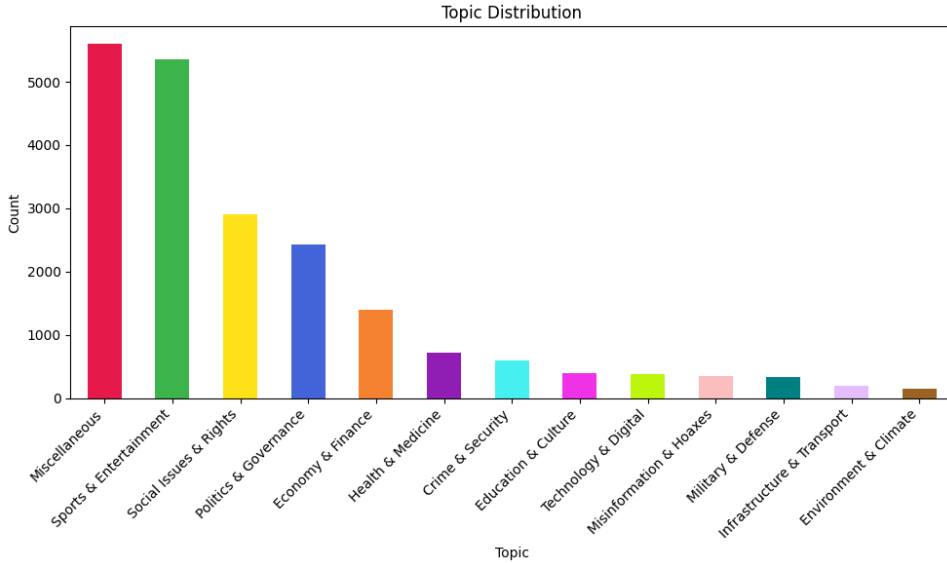


Figure 1: Distribution of topics across the dataset

3.2 LLM-Augmented Data Generation

To mitigate the underrepresentation of specific content types and dialectal variations, we leveraged *Claude Sonnet 4.5* and *Gemini 3 Thinking*, chosen for their demonstrated proficiency in Algerian Darija and code-switching phenomena. These models were employed in two distinct augmentation settings:

- **MSA-to-Darija Translation:** In the first setting, the models were used to translate instances originally written in Modern Standard Arabic (MSA) into Algerian Darija (different underrepresented forms were considered—e.g., Darija text with latin scripts). Each translated instance inherited the class label of its corresponding original sample. Carefully designed prompts were provided to ensure that translations remained strictly within Algerian Darija, avoiding regression to MSA or drift toward other Maghrebi dialects, particularly Moroccan Darija.
- **Fictional Instance Generation:** In the second setting, models generated fictional content in Algerian Darija, labeled as False (F). To ensure high-quality synthetic data, the prompts were designed to reflect the linguistic patterns identified during our initial analysis of the collected fake news dataset. Furthermore, we explicitly instructed the models to avoid predictable patterns and maintained a strict "clean room" approach to prevent data leakage from the source materials. This ensured the output remained authentically dialectal while being stylistically diverse.

In both settings, seed examples derived from organically collected data were incorporated into the prompts to guide linguistic style and class conformity. All generated and translated instances underwent the same annotation pipeline and manual quality assurance procedures as organic data, ensuring consistency, label reliability, and dialectal authenticity across the dataset.

3.3 Annotation Framework

3.3.1 Annotation Taxonomy

We adopted a taxonomy reflecting misinformation’s nuanced spectrum:

- **Real (R):** Verifiable facts
- **Misleading (M):** Factual elements distorted through decontextualization or selective omission
- **False (F):** Entirely fabricated content
- **Satire (S):** Intentional news mimicry for humor
- **Non-news (N):** Content lacking factual claims

3.3.2 Annotation Protocol and Label Adjudication

Initially, each instance received independent labels from three annotators, including the original collector to preserve contextual understanding. Final labels were determined through a structured adjudication process, and instances with complete, unresolvable disagreement were excluded, as such cases were interpreted as genuine epistemic ambiguity rather than annotation noise.

First Annotation Iteration. In the first iteration, data collected from all sources were pooled and assigned to a three-annotator scheme. Each instance was labeled by a randomly selected triplet from a team of six annotators, resulting in varying annotator combinations rather than a fixed panel. This design enabled the computation of agreement measures both at the triplet level and across the pooled dataset, providing a robust assessment of labeling consistency while reducing systematic bias tied to specific annotator groups. Annotators worked independently and were provided with precise definitions for each label to ensure consistent interpretation.

Subsequent Iterations. In later data collection iterations, annotation was predominantly performed by a single annotator due to time and scalability constraints. This transition was informed by the high inter-annotator agreement observed during the initial phase, allowing for faster dataset expansion while maintaining acceptable labeling reliability. Annotators were free to choose their verification methodology, commonly relying on cross-referencing reputable news outlets, timeline alignment for temporal consistency, historical fact-checking, and assessment of source credibility. When appropriate, automated labeling strategies—including rule-based heuristics and large language model assistance—were employed to accelerate the process, with human oversight retained for ambiguous or high-impact cases.

Label Adjudication and Conflict Resolution. Final label selection followed a strict priority-based rule hierarchy designed to favor conservative handling of misinformation and transparent treatment of uncertainty. Unanimous agreement resulted in direct label acceptance, reflecting maximal confidence. In cases where the majority classified an instance as N (Not News) but a minority assigned F, M, R, or S, the minority label was preserved to prioritize potentially informative content. Clear majorities were otherwise accepted, while fundamental truth-value conflicts (e.g., F vs. R) were marked as T (Tie) to avoid forced resolution.

Ties involving M (Misleading) were conservatively resolved in favor of M when opposed to either R or F, reflecting the higher risk associated with partially true yet deceptive content. When no prior rule applied, the label assigned by the original collector was used to leverage contextual familiarity with the source. Instances with only a single annotation retained that label to preserve coverage, whereas items lacking annotations or exhibiting unresolved ties without collector involvement were marked as T and excluded, ensuring transparency and dataset integrity.

3.4 Data Preprocessing

To ensure consistency, reliability, and suitability for downstream misinformation detection tasks, the dataset underwent a multi-stage preprocessing pipeline combining normalization, semantic consolidation, content cleaning, and quality flagging. All preprocessing steps were designed to be non-destructive, preserving original data while enabling controlled filtering during analysis and model training.

3.4.1 Topic Normalization via Semantic Remapping

The original sources employed heterogeneous topic labels with varying granularity and naming conventions. To address this, we consolidated all topic annotations into a unified

taxonomy of twelve high-level categories covering political, social, economic, technological, and informational domains, including a dedicated category for misinformation and hoaxes. Rather than relying on rule-based mappings, semantic similarity models were used to align original topic labels with the standardized taxonomy. Topic names and descriptive examples were embedded and compared to ensure semantic matching beyond surface-level keyword overlap.

3.4.2 Content Length and Structural Filtering

Items with insufficient textual content were identified and flagged based on minimal length requirements necessary for meaningful news interpretation and misinformation assessment. Rather than removing such items, they were marked for exclusion to preserve transparency and allow future re-evaluation. Missing, empty, or malformed content was handled consistently under the same criteria, preventing silent data loss.

3.4.3 Temporal and Source Iteration Normalization

All date fields were normalized into a consistent calendar format to ensure chronological comparability across sources and collection phases. Partial or underspecified dates were resolved using conservative defaults to preserve temporal ordering without fabricating precision. Additionally, a dataset iteration indicator was added to all items to enable longitudinal tracking of dataset evolution and future extensions.

3.4.4 Post-Merge Data Cleaning and Quality Control

After merging all sources, a comprehensive cleaning and validation phase was applied. The following operations were performed uniformly across the dataset:

- Standardization of label formats to resolve capitalization and naming inconsistencies.
- Manual resolution of rare inter-annotator conflicts identified during early annotation iterations.
- Flagging of low-quality OCR outputs, particularly from social media sources, due to excessive noise.
- Normalization of excessive character and punctuation repetition while preserving expressive emphasis.
- Removal of URLs, hashtags, user mentions, emojis, and invisible directionality markers, with their presence retained as auxiliary signals.
- Normalization of Arabic letter variants and quotation marks to reduce lexical sparsity.
- Removal of non-standard or control characters while preserving meaningful punctuation.
- Whitespace normalization to eliminate formatting artifacts.

- Extraction of structural and stylistic indicators (e.g., length, punctuation density, capitalization patterns) prior to cleaning.
- Flagging of low-quality items based on insufficient lexical content, abnormal character composition, or spam-like repetition patterns.

While elements were removed from the textual content to reduce noise, their presence was explicitly preserved through a set of auxiliary metadata features. These features captured properties of the original text prior to cleaning, including content length, URL presence, hashtag and mention frequency, emoji usage, capitalization patterns, punctuation density, exclamation and question mark usage, and character composition indicators. This design allowed the dataset to retain stylistic and structural signals that may be informative for misinformation detection while preventing direct lexical leakage.

These metadata features were later used both independently and in combination with textual embeddings during ablation studies, enabling a controlled evaluation of their individual and complementary contributions to model performance.

3.4.5 Illustrative Examples of Cleaning and Flagging

Table 1 presents representative Arabic examples illustrating the types of content modifications and flags applied during preprocessing.

Operation	Example (Arabic)	Rationale
Character repetition normalization	هذا → هذا خبر كااااااذب!!! خبر كاااذب!!!	Reduces spam-like exaggeration while preserving emphasis
URL removal	اقرأ التفاصيل هنا https://example.com	Links add noise; presence retained as auxiliary signal
Hashtag removal	أخبار عاجلة #الجزائر #سياسة	Hashtags are metadata, not semantic content
Mention removal	@المصدر الرسمي	User references are irrelevant for content understanding
Arabic letter normalization	إعلان هام → إعلان هام جداً جداً	Reduces lexical sparsity in Arabic text
Low-content flagging	عاجل	Insufficient context for reliable classification
Low letter-ratio flagging	!!! \$\$\$ 12345 !!!	Indicative of corrupted OCR or spam content

Table 1: Examples of content cleaning and quality flagging applied during preprocessing.

3.5 Data Leakage Prevention

To ensure reliable evaluation and avoid inflated performance estimates, we adopted a strict data leakage prevention strategy guided by the principle that samples sharing nearly identical semantic or structural identity must never cross the train, validation, and test splits. Given the prevalence of reposts, recurring events, and rich metadata in the dataset, leakage prevention was enforced through a multi-stage pipeline applied prior to model training.

3.5.1 Pre-Split Filtering

Before any dataset partitioning, a set of hard exclusions was applied:

- Items explicitly marked as duplicates were removed to eliminate redundant representations.
- Items with no remaining linguistic content after cleaning were excluded, as they provide no usable signal for text-based modeling.
- Instances assigned the tie label (T), representing unresolved epistemic uncertainty, were excluded from supervised splits to ensure label reliability.

These steps reduce noise and prevent ambiguous or degenerate samples from contaminating model training and evaluation.

3.5.2 Exact Deduplication

All remaining samples were subjected to exact deduplication using normalized text representations. Samples sharing identical normalized content were collapsed into a single canonical instance. This step ensures that verbatim reposts or copied captions do not appear across different dataset splits, preserving the integrity of evaluation metrics.

3.5.3 Near-Duplicate and Event-Based Grouping

To address semantic overlap beyond exact text matches, samples were clustered into groups representing near-duplicates or shared real-world events. Grouping was based on semantic similarity computed from textual representations, reinforced with contextual metadata such as original topic assignments and normalized source identifiers. Each sample was assigned a group identifier, with all semantically equivalent or event-related instances sharing the same group.

This grouping mechanism captures reposts, paraphrases, and cross-platform discussions of the same event, preventing memorization effects during training.

3.5.4 Split Construction with Group-Level Constraints

Dataset splitting was performed under strict group-aware constraints:

- No group identifier was allowed to appear in more than one split.
- Label distributions were preserved at the group level rather than the instance level.
- Upper bounds were enforced on semantic similarity between training and test samples to limit cross-split semantic overlap.

Applying these constraints ensures that evaluation reflects genuine generalization to unseen content rather than recognition of near-identical samples or repeated events.

3.6 Corpus Characteristics

3.6.1 Dataset Composition

The final corpus contains 22,391 instances distributed across training (15,699, 70.1%), validation (3,348, 15.0%), and test (3,344, 14.9%) sets with preserved class distribution (Table 2). A comprehensive breakdown of the linguistic, structural, and annotation-level statistics for the Algerian Dialect Corpus—including vocabulary richness, and punctuation density is provided in Appendix 6.3.

Table 2: Class distribution across data splits.

Split	False	Real	Non-news	Misleading	Satire	Total
Train	4,779 (30.44%)	3,896 (24.82%)	3,830 (24.40%)	2,869 (18.28%)	325 (2.07%)	15,699
Validation	965 (28.82%)	851 (25.42%)	828 (24.73%)	641 (19.15%)	63 (1.88%)	3,348
Test	953 (28.50%)	849 (25.39%)	872 (26.08%)	592 (17.70%)	78 (2.33%)	3,344
Total	6,697 (29.91%)	5,596 (24.99%)	5,530 (24.70%)	4,102 (18.32%)	466 (2.08%)	22,391

As illustrated in Figure 2, the dataset exhibits a moderate class imbalance across most labeling categories. The distribution is led by False (F), Non-news (N), and True (R) content, which constitute the majority of the samples. In contrast, Satire (S) represents the smallest fraction of the dataset reflecting a severe imbalance, accounting for approximately 2.2% of the total entries. To ensure the model remains robust against this natural skew, class weighting was applied during training.

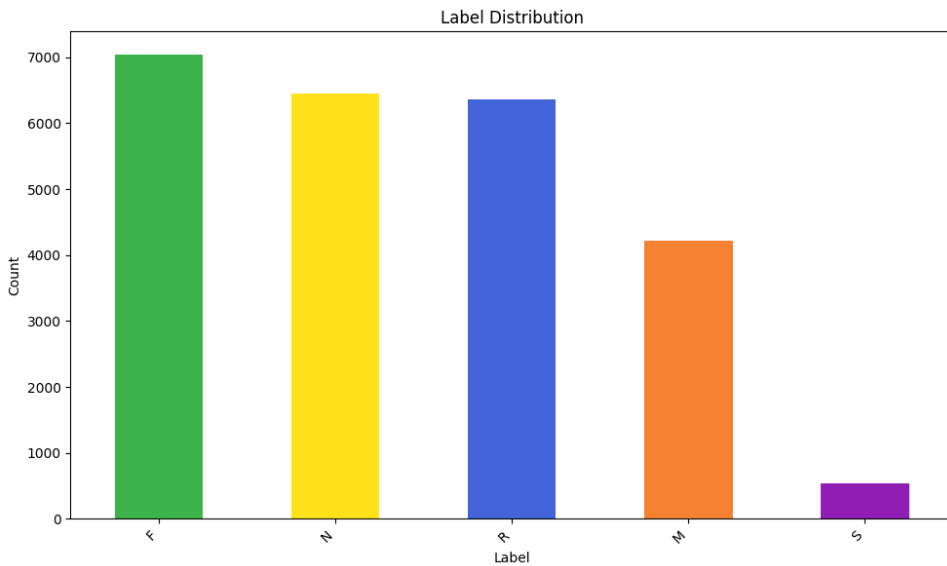


Figure 2: Distribution of labels across the entire dataset.

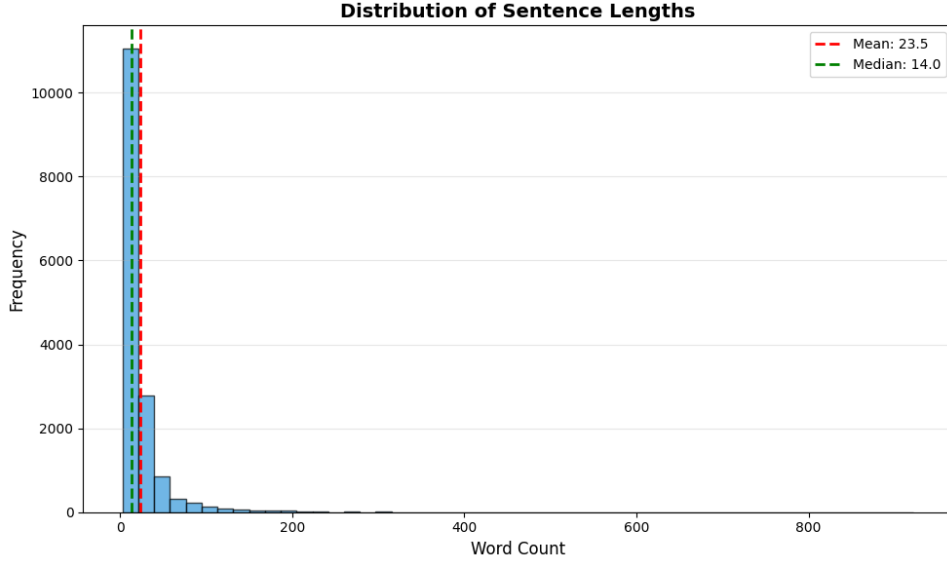


Figure 3: Distribution of sentence length across the training set.

3.6.2 Topic Coverage

Content spans miscellaneous interests, sports and entertainment, social issues and rights, politics and governance, and additional topical domains to ensure broad representativeness, as illustrated in Figure 1.

3.6.3 Language Varieties

The dataset encompasses Algerian Darija (various regional variations), Modern Standard Arabic (MSA), French code-switching, Arabizi (Latin-script Arabic), and mixed linguistic forms reflecting natural online conversational news reporting styles.

3.6.4 Data Sources

The dataset was collected from a combination of social media and news platforms, with sources spanning human-authored, AI-translated, and AI-generated content.

Social Media: Facebook (official and non-official pages, user posts), YouTube (video descriptions and comments), Instagram (posts and captions), TikTok (comments and description), WhatsApp and X. These platforms provide rich, real-world examples of user-generated content and event-specific discussions.

News Platforms: Official and non-official media outlets, pages disseminating misleading content, and anti-misinformation reporting sources. These sources contribute more formal news and verified reports, complementing social media content.

AI-Involved Content: AI-generated and AI-translated samples were explicitly included to inject more variety in topics and language styles.

Figure 4 shows the distribution of the dataset across these source categories. YouTube contributes the largest portion, followed by AI-generated content and Facebook posts, while Instagram, AI-translated content, web pages, and other miscellaneous sources comprise smaller fractions of the dataset. This distribution reflects both the prevalence of social media activity and the integration of synthetic content.

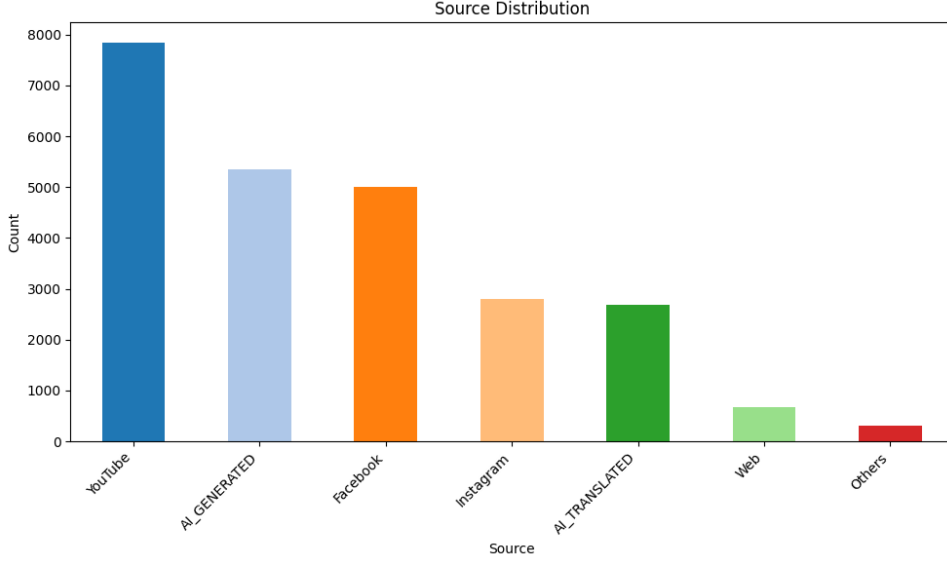


Figure 4: Distribution of dataset items by source.

3.7 Inter-Annotator Agreement

To assess annotation reliability during dataset construction, we conducted an inter-annotator agreement (IAA) analysis on the initial multi-annotator phase. Two deliberate simplifications were applied for this analysis. First, items labeled as **N** (Not News) were excluded, as they do not represent claims requiring factual assessment. Second, due to consistently high confusion between the **M** (Misleading) and **F** (False) labels, these two categories were merged for the purpose of agreement computation. This simplification provides insight into upper-bound agreement and may serve as a reference point for future work exploring finer-grained distinctions between misleading and false content.

Annotation Setup. The agreement analysis was conducted on a sample of 400 items, each annotated independently by three annotators. Importantly, annotations were produced by a pool of six annotators, resulting in multiple annotator triplets rather than a fixed set of three annotators across all items. In total, eleven unique annotator triplets were used. This setup reflects realistic annotation conditions and reduces systematic bias associated with fixed panels, while still enabling reliable agreement estimation.

After normalization, the effective label set consisted of three categories: **M/F** (merged Misleading and False), **R** (Real/Reporting), and **S** (Satire).

Observed Agreement. Following label normalization, perfect agreement among all three annotators was observed for 78.85% of items, while the remaining 21.15% exhibited partial agreement (two annotators agreeing). No instances of complete disagreement were observed, indicating a strong shared understanding of label semantics at the coarse-grained level.

Fleiss’ Kappa. To measure chance-corrected agreement among multiple annotators, we report Fleiss’ Kappa (κ), defined as:

$$\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$$

where $\bar{P}$ denotes the observed agreement across items and $\bar{P}_e$ the agreement expected by chance given the marginal label distribution.

In our setting, Fleiss’ Kappa reached a value of $\kappa = 0.7322$, with an observed agreement $\bar{P} = 0.8590$ and chance agreement $\bar{P}_e = 0.4734$. According to standard interpretation guidelines, this corresponds to *substantial agreement*, indicating strong consistency among annotators beyond chance.

Krippendorff’s Alpha. To complement Fleiss’ Kappa, we also report Krippendorff’s Alpha (α), which is robust to varying annotator combinations and explicitly models disagreement. It is defined as:

$$\alpha = 1 - \frac{D_o}{D_e}$$

where D_o represents observed disagreement and D_e the disagreement expected by chance.

The resulting Krippendorff’s Alpha was $\alpha = 0.5390$, with $D_o = 0.2472$ and $D_e = 0.5362$. While this value falls below commonly cited acceptability thresholds for strong reliability, it reflects the inherent subjectivity of misinformation-related judgments, particularly when distinguishing nuanced cases even after merging M and F. The use of multiple annotator triplets further increases variability, making this estimate a conservative assessment of agreement.

3.8 Comparative Analysis and Contributions

Table 3 positions our work against related efforts. Unlike FASSILA and BOUTEF, which adopt binary classification schemes, our five-class taxonomy captures the nuanced spectrum of misinformation. While AFND provides large-scale coverage of Modern Standard Arabic news content, our dataset focuses on dialectal social media data, where misinformation predominantly circulates in the Algerian context.

Table 3: Comparison with existing Arabic dialectal misinformation datasets.

Dataset	Size	Classes	Dialect	Domain
FASSILA	10,087	2 (Binary)	Algerian	Multi-domain
BOUTEF	3,600	2 (Binary)	Algerian/Tunisian	Social Media
AFND	600,000+	2 (Binary)	MSA	News Agencies
Ours	22,391	5 (Multi-class)	Algerian Darija	Social Media + News

3.9 Limitations

Our dataset faces notable constraints. Severe Satire underrepresentation (2.08% overall) stems from sourcing challenges—formal satirical content in news-mimicking format proved difficult to collect adequately within our timeframe. Additionally, LLM-generated instances used to address content gaps may introduce subtle distributional shifts from naturally occurring social media discourse.

4 Results and Analysis

This section presents a comprehensive evaluation of misinformation detection models trained on Algerian Darija text. We evaluate eight models across three architectural

paradigms: classical machine learning (Logistic Regression and Linear SVM), neural networks (CNN and BiLSTM), and transformer-based models (DziriBERT, XLM-RoBERTa-base, mBERT, and XLM-RoBERTa-large). All models were evaluated on the same test set of 3,344 samples to ensure fair comparison.

4.1 Evaluation Methodology

4.1.1 Baseline Models

To establish comprehensive performance benchmarks, we employed six baseline models spanning classical machine learning, neural sequence architectures, and transformer-based approaches:

- **TF-IDF + Logistic Regression:** Multinomial logistic regression with L2 regularization operating on TF-IDF features (up to 10,000 unigram-trigram features). Class weights inversely proportional to frequencies address class imbalance.
- **TF-IDF + Linear SVM:** Linear support vector machine with squared hinge loss using identical TF-IDF representations. Provides interpretable baseline for assessing lexical separability.
- **CNN:** Convolutional neural network with multiple one-dimensional filters of varying kernel sizes, max-pooling, and dropout regularization. Uses pretrained AraVec embeddings to initialize the embedding layer, capturing local patterns while maintaining computational efficiency.
- **BiLSTM:** Bidirectional LSTM processing sequences in both directions to capture long-range dependencies and contextual interactions in dialectal and code-switched text.
- **DziriBERT:** Dialect-specific BERT-base model (12 layers, 12 attention heads) pretrained on Algerian Darija corpora. Tests the hypothesis that dialect-specific pretraining improves low-resource language representation.
- **XLM-RoBERTa-base:** Large-scale multilingual transformer pretrained on 100+ languages with byte-pair encoding tokenizer. Handles code-switching and mixed-script inputs prevalent in Algerian online discourse.
- **mBERT:** Multilingual BERT with shared WordPiece vocabulary across 100+ languages. Serves as intermediate design point between dialect-specific and large-scale multilingual models.
- **XLM-RoBERTa-Large:** High-capacity multilingual transformer (550M parameters) with large BPE vocabulary. Fine-tuned with reduced batch size, gradient accumulation, and targeted augmentation (5% minority-class oversampling) to address severe class imbalance. Provides upper-bound reference for transformer performance despite higher computational cost.

4.1.2 Evaluation Metrics

Our evaluation follows the specifications outlined in Section 4, using metrics designed to address the severe class imbalance in our dataset. Our **primary metric** is the *macro-averaged F1-score*, which treats all classes equally regardless of support size, ensuring minority classes receive appropriate evaluation weight.

We report the following metrics for each model:

- **Macro-averaged:** Precision, Recall, and F1-score (unweighted average across classes)
- **Weighted-averaged:** Precision, Recall, and F1-score (weighted by class support)
- **Per-class:** Precision, Recall, and F1-score for all five classes
- **Overall accuracy:** Reported for reference but not used as primary metric

All models employed class weighting during training, with the Satire class receiving a weight 9.66 times higher than other classes. The test set was evaluated only once after training completion to prevent data leakage.

4.2 Dataset Partitioning

To facilitate the training and evaluation of our models, we partitioned the Algerian Dialect Corpus into three distinct subsets: training, validation, and test. The final counts for each split are as follows:

- **Train:** 15,699 samples
- **Validation:** 3,348 samples
- **Test:** 3,344 samples

The consistency of the label distribution across these splits—ensuring that the class imbalance (particularly for the Satire class) is proportionally represented in the training and evaluation phases—is visualized in Figure 5.

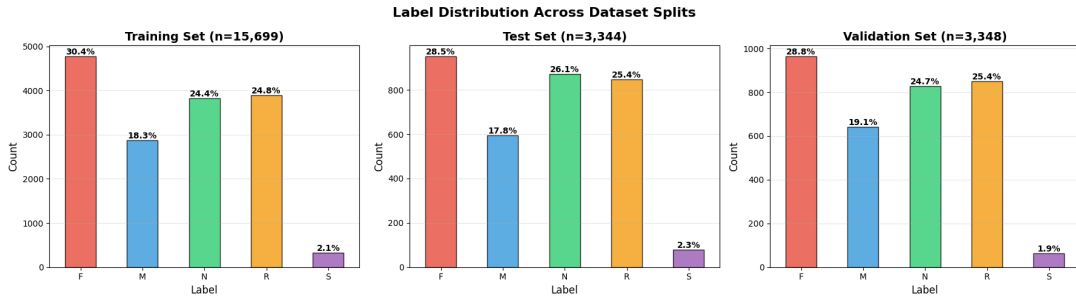


Figure 5: Label distribution across the training, validation, and test sets.

Table 4: Test set performance comparison. Best macro F1-score (primary metric) in bold.

Model	Acc	Macro P	Macro R	Macro F1	Wtd F1
<i>Classical Models</i>					
Logistic Regression	0.729	0.643	0.666	0.647	0.737
Linear SVM	0.717	0.618	0.614	0.615	0.718
<i>Neural Networks</i>					
CNN	0.704	0.629	0.653	0.617	0.728
BiLSTM	0.694	0.619	0.623	0.602	0.719
<i>Transformer Models</i>					
DziriBERT	0.773	0.685	0.669	0.675	0.772
XLM-RoBERTa-base	0.762	0.675	0.679	0.673	0.768
mBERT	0.754	0.650	0.644	0.645	0.757
XLM-R-large + Aug	0.783	0.711	0.674	0.682	0.784

4.3 Performance Results

Table 4 presents the complete test set performance across all models.

The results reveal clear performance hierarchies. Transformer models significantly outperform both classical and neural approaches, with XLM-RoBERTa-large achieving the highest macro F1-score of 0.682, representing a 5.4% improvement over the best classical model (Logistic Regression, 0.647). Among transformers, DziriBERT achieves the second-best performance (0.675), demonstrating the value of domain-specific pretraining on Algerian Darija. Classical models outperform neural networks, suggesting that for this task, traditional feature engineering with TF-IDF can be more effective than basic neural architectures without pretrained embeddings.

4.4 Per-Class Performance Analysis

Table 5 presents detailed per-class F1-scores for selected models, revealing substantial variation across misinformation types.

Table 5: Per-class F1-scores on test set. Best score per class in bold.

Model	F (Fake)	R (Real)	N (Non-news)	M (Misleading)	S (Satire)
Logistic Reg.	0.825	0.601	0.773	0.738	0.301
Linear SVM	0.816	0.562	0.772	0.706	0.219
CNN	0.824	0.724	0.775	0.584	0.178
BiLSTM	0.810	0.729	0.765	0.571	0.133
DziriBERT	0.848	0.770	0.841	0.615	0.301
XLM-R-base	0.842	0.777	0.807	0.640	0.301
mBERT	0.837	0.764	0.810	0.615	0.199
XLM-R-large	0.850	0.804	0.832	0.646	0.278

Majority Classes (F, R, N): Transformer models excel on these classes, with F1-scores consistently above 0.75. XLM-RoBERTa-large achieves the best Fake detection (0.850) and Real content classification (0.804), while DziriBERT performs best on Non-news

(0.841), benefiting from its Algerian dialect pretraining. These classes demonstrate that transformers can reliably identify content with clear linguistic patterns.

Misleading Class: This category proves moderately challenging, with F1-scores ranging from 0.571 (BiLSTM) to 0.738 (Logistic Regression). Interestingly, the classical Logistic Regression model achieves the best Misleading detection overall, outperforming all transformer models. Among transformers, XLM-RoBERTa-large achieves the best score (0.646). The relatively lower transformer performance suggests that Misleading content’s ambiguous nature—blending factual and non-factual elements—may be better captured by traditional feature engineering than by contextualized representations alone.

Satire Class: Satire detection represents the primary challenge across all architectures. XLM-RoBERTa-base achieves the best performance at F1=0.301, marginally outperforming DziriBERT (0.301) and Logistic Regression (0.301), while neural networks perform substantially worse (BiLSTM: 0.133, CNN: 0.178). The severe performance degradation stems from: (1) extreme class imbalance with only 78 test samples (2.3%), (2) satire’s subtle linguistic nature requiring cultural understanding, and (3) confusion with other content types. Notably, XLM-RoBERTa-large’s Satire F1-score is 0.278, lower than smaller models despite data augmentation increasing training representation from 2.3% to 4.8%, suggesting that quantity alone cannot overcome the inherent difficulty of this class.

4.5 Metadata Features Ablation Study

We evaluated the impact of metadata features by comparing baseline (text-only) and enhanced (text+metadata) variants of classical models.

Table 6: Ablation study on validation and test sets showing metadata feature impact.

Model	Variant	Set	Macro F1	Satire F1	Change
2*Logistic Reg.	Baseline	Val	0.642	0.193	–
	+ Metadata	Val	0.655	0.267	+2.1%
	Baseline	Test	–	–	–
	+ Metadata	Test	0.647	0.301	–
2*Linear SVM	Baseline	Val	0.634	0.233	–
	+ Metadata	Val	0.640	0.262	+0.9%
	Baseline	Test	–	–	–
	+ Metadata	Test	0.615	0.219	–

Metadata features provide minimal improvements. On validation data, Logistic Regression gains 2.1% in macro F1-score, while Linear SVM improves by only 0.9%. The most notable impact appears in Satire detection on validation (Logistic Regression: 0.193→0.267), though test set generalization varies. These results indicate that textual content carries the primary discriminative signal for Algerian Darija misinformation detection, with surface-level features contributing marginally.

4.6 Error Analysis and Confusion Patterns

Analysis of confusion matrices (see Appendix A) reveals systematic misclassification patterns across models. We focus on DziriBERT and XLM-RoBERTa-large as representative examples.

Table 7: Most common confusion patterns for DziriBERT and XLM-RoBERTa-base.

Model	Count	Confusion Pattern
5*DziriBERT	99	Real → Misleading
	89	Fake → Misleading
	84	Misleading → Fake
	73	Misleading → Real
	61	Non-news → Misleading
5*XLM-R-base	139	Fake → Misleading
	95	Real → Misleading
	86	Non-news → Misleading
	77	Non-news → Real
	70	Misleading → Real

Misleading as confusion hub: The most prevalent error pattern involves misclassifying content as Misleading. Both models frequently predict Misleading for Real, Fake, and Non-news samples. This bidirectional confusion (e.g., Real→Misleading and Misleading→Real) suggests that Misleading content shares linguistic features with multiple classes, making boundaries difficult to distinguish.

Satire dispersal: Satire samples scatter across predictions. The normalized confusion matrices (see Appendix A) show that Satire correct identification rates vary substantially: XLM-RoBERTa-base achieves 32%, DziriBERT 26%, while mBERT manages only 18%. This scattered distribution reflects the fundamental difficulty in detecting satirical intent, which often mimics legitimate news formats while presenting absurd content.

Strong diagonal for majority classes: Despite confusion in challenging cases, both models maintain strong correct classification rates for Fake (larger than 80%), Real (larger than 75%), and Non-news (larger than 79%), demonstrating reliable performance on well-represented classes.

4.7 Training Dynamics

Training curves for transformer models (see Appendix A) reveal consistent convergence patterns. All models achieve rapid learning in epochs 1–2, with validation macro F1-scores improving from ~ 0.59 – 0.61 to ~ 0.64 – 0.68 . Best performance occurs at epoch 3 for most models:

Training loss consistently decreases from ~ 1.15 – 1.18 to ~ 0.32 – 0.50 , while validation loss shows characteristic U-shaped curves: initial decrease followed by increase after epoch 3–4. This divergence indicates mild overfitting in later epochs, validating our early stopping strategy with patience=5. XLM-RoBERTa-large requires substantially longer training (82 minutes) due to its 355M parameters and data augmentation overhead, compared to 22–24 minutes for base models.

Table 8: Best validation performance and convergence characteristics.

Model	Best Epoch	Best Val F1	Training Time
DziriBERT	2	0.708	21.97 min
XLNet-RoBERTa-base	5	0.685	23.99 min
mBERT	3	0.680	23.11 min
XLNet-RoBERTa-large	3	0.692	81.88 min

4.8 Key Findings

Our evaluation reveals several critical insights:

- 1. Transformer superiority:** Transformer models substantially outperform classical and neural approaches. The best transformer achieves 5.4% higher macro F1 than the best classical model, demonstrating the importance of contextualized representations for Algerian Darija misinformation detection.
- 2. Domain-specific pretraining benefits:** DziriBERT’s specialized Algerian Darija, without taking data augmentation into account, pretraining yields the highest accuracy (77.3%) and strongest performance on Fake and Non-news classes, though general multilingual models with larger capacity can achieve competitive overall performance.
- 3. Satire detection remains unsolved:** Despite varied architectures and data augmentation, no model exceeds F1=0.301 on Satire, indicating this task requires even more satire data beyond augmentation.
- 4. Classical baselines are competitive:** Logistic Regression outperforms both neural networks and achieves the best Misleading class F1-score, suggesting that well-engineered features can be surprisingly effective for this task.
- 5. Limited metadata utility:** Surface-level features contribute minimally to performance, with textual content providing the primary discriminative signal for classification decisions.

5 Discussion

This section discusses the experimental findings presented in the previous section, focusing on the implications of model performance differences, the impact of linguistic characteristics of Algerian Darija, and the limitations revealed by class-specific behavior. Rather than reiterating numerical results, we interpret the observed trends in light of model architecture, data properties, and task complexity.

5.1 Impact of Model Architecture on Dialectal Misinformation Detection

The results clearly demonstrate that model architecture plays a decisive role in Algerian Darija misinformation detection. Transformer-based models consistently outperform classical machine learning and neural sequence models, confirming the importance of contextualized representations for handling dialectal, informal, and code-switched text.

Classical TF-IDF-based models achieve competitive performance relative to neural networks, particularly Logistic Regression, which outperforms both CNN and BiLSTM

models in macro F1-score. This suggests that surface-level lexical cues are highly informative for several misinformation categories, especially *Misleading*. In contrast, neural architectures trained from scratch struggle to generalize, likely due to limited data size and the absence of pretrained semantic knowledge. This finding aligns with prior work indicating that shallow neural models often underperform well-regularized linear baselines in low-resource settings.

Transformer models provide a substantial performance gain by leveraging pretrained linguistic knowledge. Their ability to model long-range dependencies, semantic nuance, and multilingual context is especially beneficial in Algerian Darija, where non-standard spelling and frequent code-switching complicate purely lexical approaches.

5.2 Dialect-Specific versus Multilingual Pretraining

A key insight from the experiments is the trade-off between dialect specialization and multilingual coverage. DziriBERT, pretrained primarily on Algerian dialect data, achieves strong performance across majority classes and excels at identifying *Non-news* content. This indicates that dialect-aware pretraining captures stylistic and pragmatic patterns specific to Algerian online discourse.

However, large multilingual models such as XLM-RoBERTa-Large ultimately achieve the best overall macro F1-score. This suggests that increased model capacity and exposure to diverse languages provide better generalization, particularly for code-switched text combining Darija, MSA, and French. The results imply that while dialect-specific models are highly effective, large multilingual transformers can compensate for lack of specialization through scale and representational richness.

Interestingly, mBERT underperforms both DziriBERT and XLM-RoBERTa variants, indicating that not all multilingual pretraining strategies are equally effective. Models pretrained on social media or noisy web text appear better suited for this task than those trained on more formal corpora.

5.3 Class-Specific Challenges and Semantic Ambiguity

Per-class analysis reveals that performance varies substantially across misinformation categories. *Fake*, *Real*, and *Non-news* are consistently well recognized by transformer models, suggesting that these categories exhibit relatively stable and distinguishable linguistic patterns.

In contrast, *Misleading* emerges as a semantic boundary class. The frequent confusion between Misleading and both Real and Fake content reflects the inherent ambiguity of this category, which often combines factual information with selective framing or omission. The fact that Logistic Regression achieves the highest F1-score for this class suggests that explicit lexical cues and phrasing patterns may be more informative than deep semantic representations alone.

Satire remains the most challenging category across all architectures. Despite class weighting and targeted data augmentation, performance remains low. This is due to both the inherently ambiguous nature of satire, which is difficult even for human readers to identify, and the severe class imbalance that restricts the models' ability to learn reliable satirical patterns.

5.4 Effectiveness of Metadata and Auxiliary Features

The ablation study shows that metadata features provide only marginal improvements, primarily affecting validation performance and rarely transferring robustly to the test set. This indicates that the core discriminative signal for Algerian misinformation detection lies in textual content rather than contextual metadata.

While metadata slightly improves satire recognition during validation, the lack of consistent generalization suggests potential overfitting or weak correlation between metadata patterns and misinformation categories. These findings reinforce the importance of robust text modeling and suggest that future improvements should prioritize richer linguistic and semantic representations rather than additional surface-level features.

5.5 Training Dynamics and Generalization Behavior

Training dynamics across transformer models exhibit rapid convergence, with optimal performance achieved within a small number of epochs. The observed divergence between training and validation loss after early epochs confirms the necessity of early stopping, particularly in imbalanced, low-resource scenarios.

XLM-RoBERTa-Large, despite its superior performance, incurs a substantial computational cost, raising practical considerations for deployment. The relatively small performance gap between large and base transformer models suggests that DziriBERT and XLM-RoBERTa-base may represent better trade-offs between accuracy and efficiency in real-world applications.

5.6 Implications for Algerian Darija NLP

Overall, the findings highlight several broader implications. First, transformer-based models are essential for effective misinformation detection in Algerian Darija, but architectural scale alone is insufficient to address deeply pragmatic phenomena such as satire. Second, dialect-specific resources remain crucial, particularly for capturing informal and culturally grounded language use. Finally, the persistent difficulty of minority classes underscores the need for more balanced datasets and richer annotation schemes that better capture intent and discourse context.

These insights provide a foundation for future research directions, which are discussed in the following section.

6 Conclusion

This work addresses a critical gap in misinformation detection for Algerian Darija by introducing a multi-class annotated corpus and evaluating a range of baseline models for fake news detection. We constructed a dataset of 22,391 instances spanning five categories—Real, False, Misleading, Non-news, and Satire—collected from social media platforms and news sources where misinformation actively circulates in Algeria. The corpus captures the linguistic complexity of Algerian online discourse, including dialectal variations, code-switching between Arabic and French, and informal writing styles.

Our experiments with eight models across classical, neural, and transformer-based architectures reveal several key findings. Transformer models substantially outperform

traditional approaches, with XLM-RoBERTa-Large achieving the best overall macro F1-score of 0.682. DziriBERT, pretrained specifically on Algerian dialect, demonstrates strong performance on majority classes, highlighting the value of domain-specific pre-training.

The confusion between Misleading content and other categories also persists across all models, reflecting the inherent ambiguity when factual elements are presented with deceptive framing. These difficulties point to fundamental challenges in automated misinformation detection that go beyond model architecture.

The contributions of this work are threefold. First, we provide the largest multi-class Algerian Darija misinformation dataset to date, addressing a critical resource gap for this low-resource dialect. Second, we establish comprehensive baselines across diverse architectures, offering benchmarks for future research. Third, our analysis identifies specific challenges particularly Satire detection and Misleading content disambiguation that require further investigation.

By releasing this corpus and establishing performance baselines, we hope to enable further research on Algerian Darija misinformation detection and contribute to building more trustworthy information ecosystems in Arabic-speaking communities.

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Appendix A

This appendix provides complementary material to support the experimental analysis presented in the the paper. It includes detailed performance metrics in the form of confusion matrices for all evaluated models, training history plots illustrating the optimization dynamics of transformer-based architectures, and representative graphical user interface screenshots demonstrating the system’s predictions on real examples.

6.1 Performance Metrics

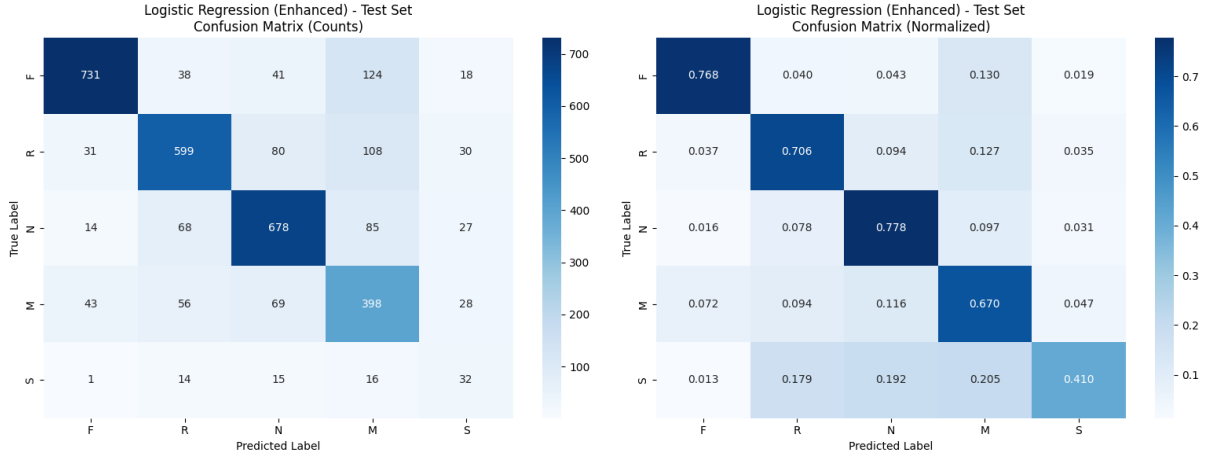


Figure 6: Confusion matrix for Model 1.1 (TF-IDF + Logistic Regression) on the test dataset.

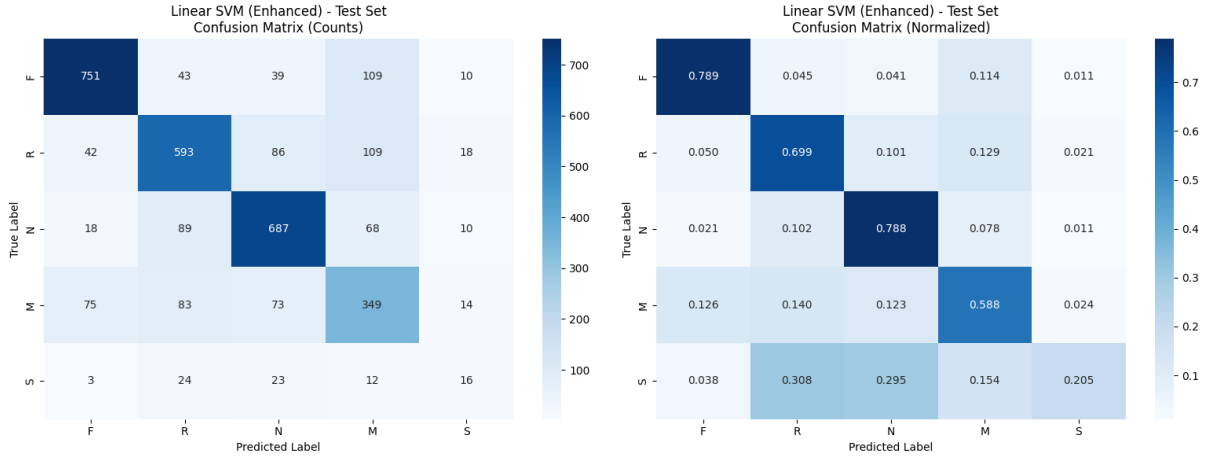


Figure 7: Confusion matrix for Model 1.2 (TF-IDF + Linear SVM) on the test dataset.

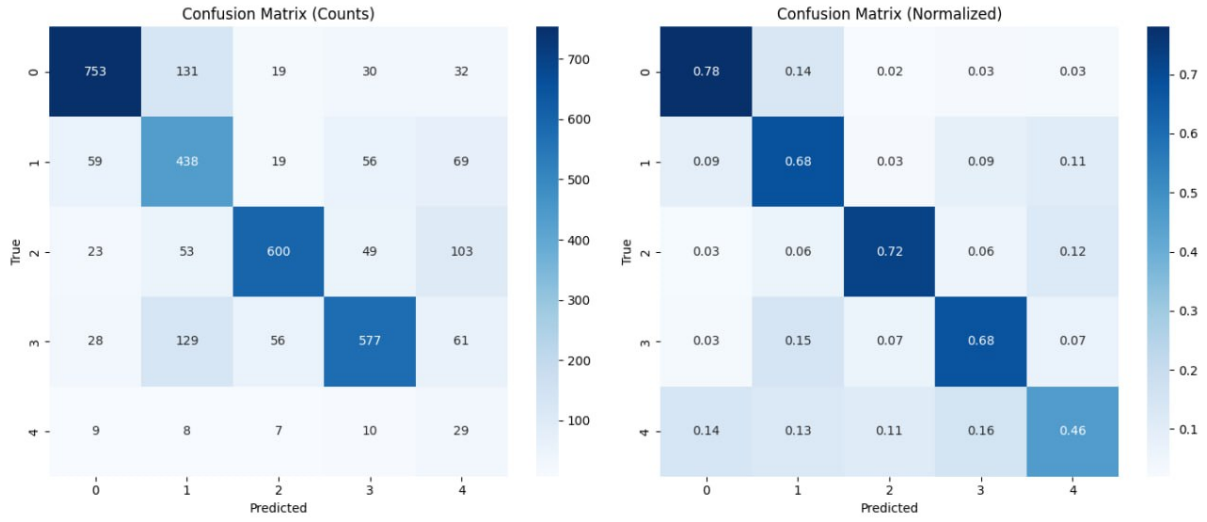


Figure 8: Confusion matrix for Model 2.1 (CNN + Word Embeddings) on the test dataset.

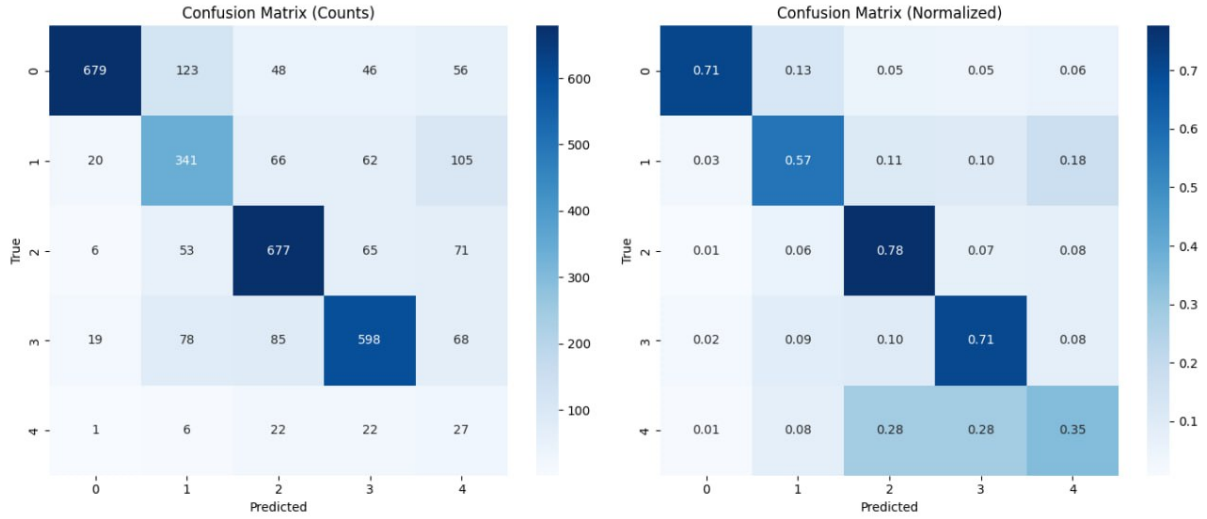


Figure 9: Confusion matrix for Model 2.2 (Bidirectional LSTM) on the test dataset.

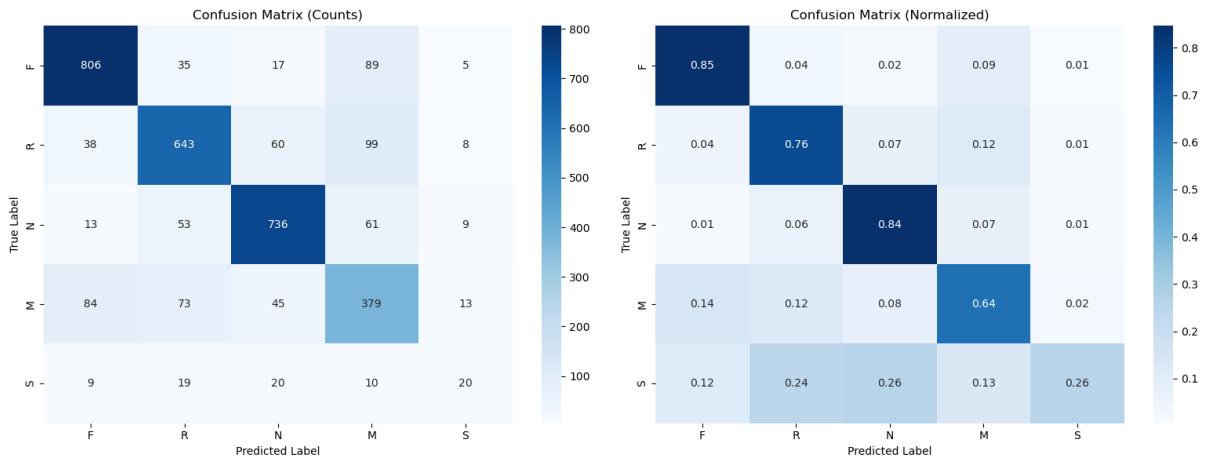


Figure 10: Confusion matrix for Model 3.1 (DziriBERT (Fine-tuned)) on the test dataset.

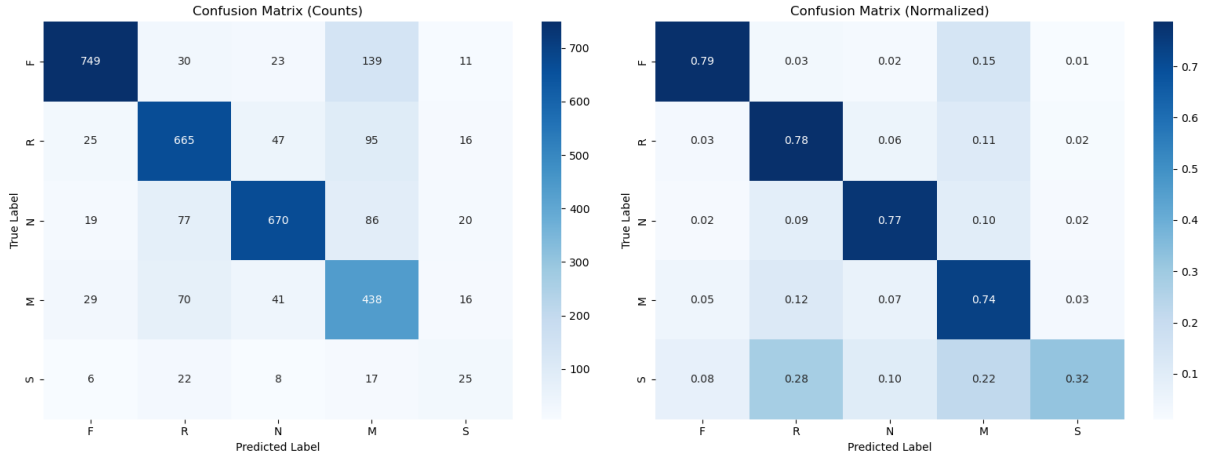


Figure 11: Confusion matrix for Model 3.2 (XLM-RoBERTa-base (Fine-tuned)) on the test dataset.

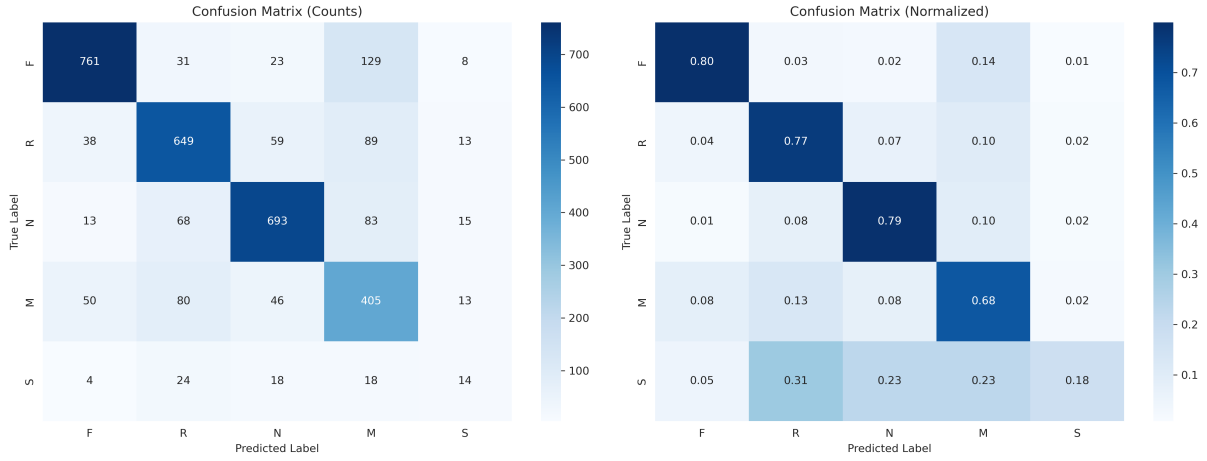


Figure 12: Confusion matrix for Model 4.1 (mBERT (Fine-tuned)) on the test dataset.

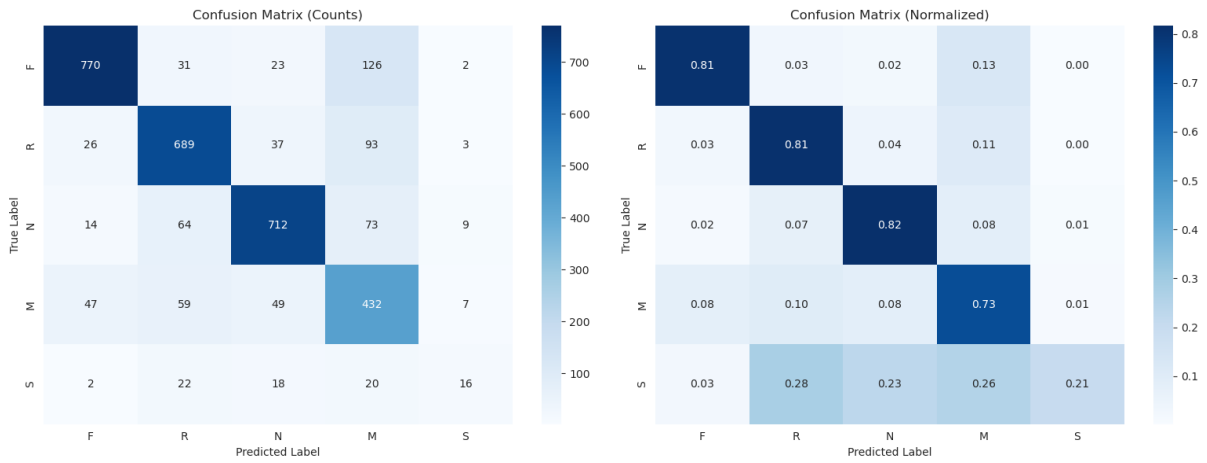


Figure 13: Confusion matrix for Model 4.2 (XLM-RoBERTa-Large) on the test dataset.

6.2 Training Plots

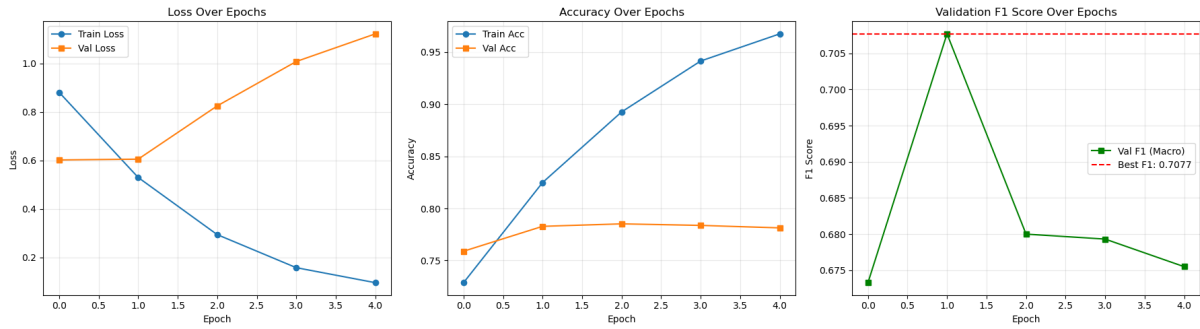


Figure 14: Training history plots for Model 3.1 (DziriBERT (Fine-tuned)).

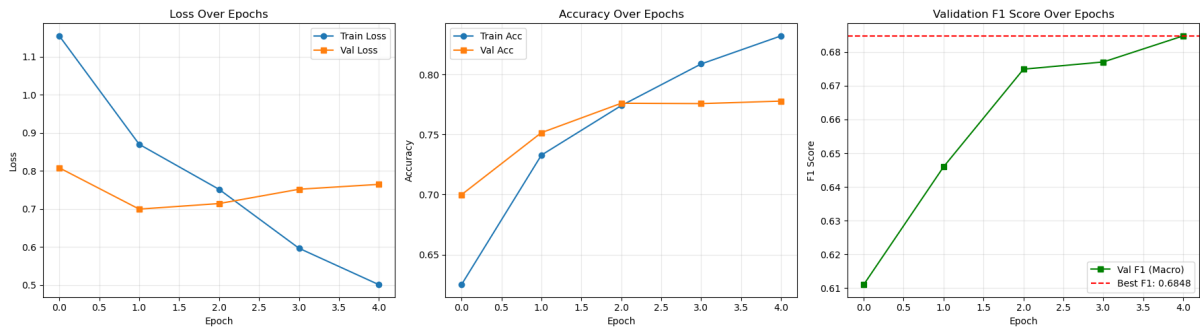


Figure 15: Training history plots for Model 3.2 (XLM-RoBERTa-base (Fine-tuned)).

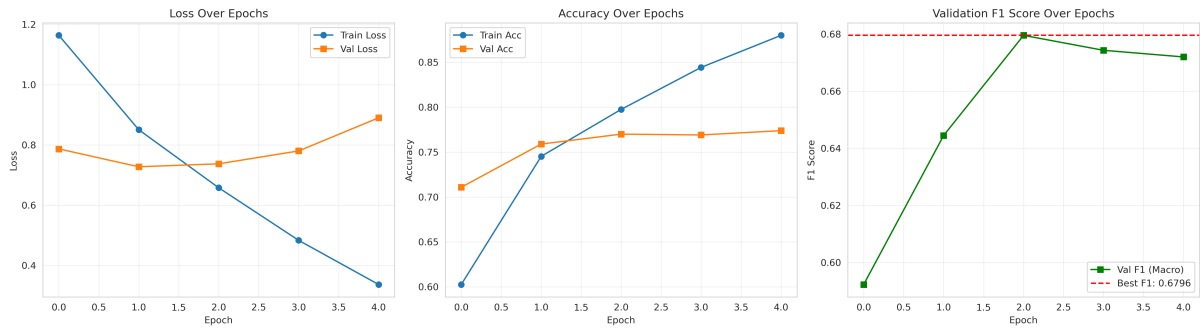


Figure 16: Training history plots for Model 4.1 (mBERT (Fine-tuned)).

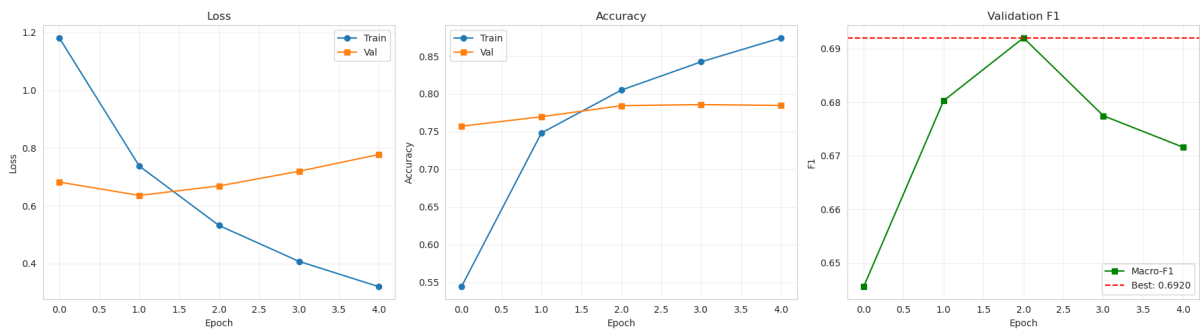


Figure 17: Training history plots for Model 4.2 (XLM-RoBERTa-Large).

6.3 GUI Screenshots

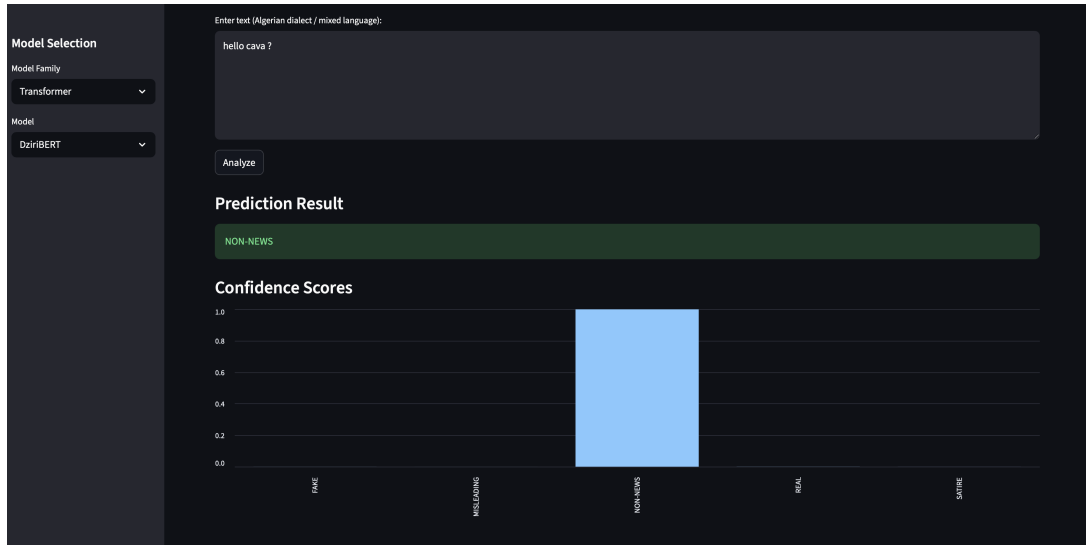


Figure 18: Screenshot demonstrating the result of our system on a sample example (human annotation "NON-NEWS").

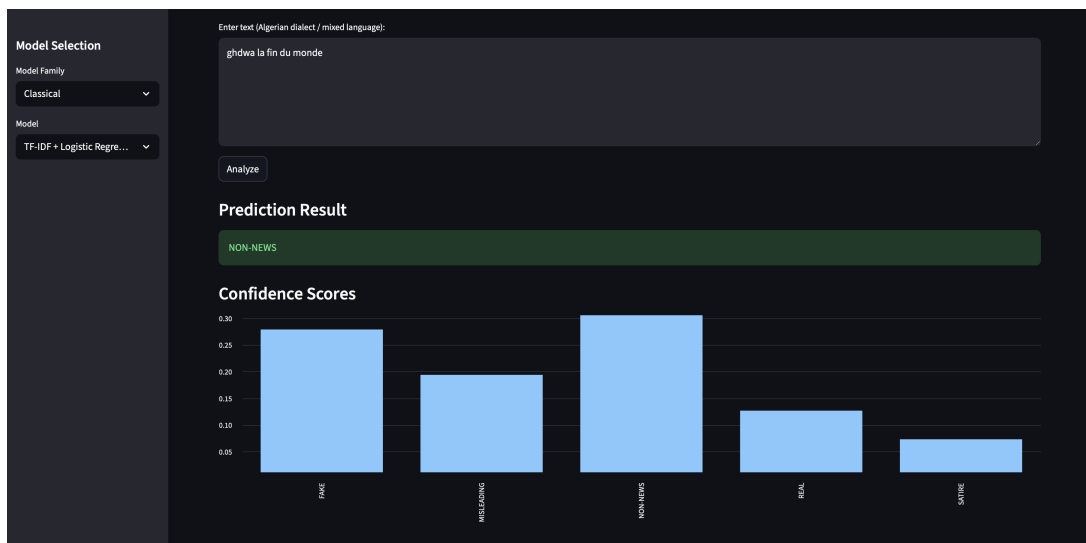


Figure 19: Screenshot demonstrating the result of our system on a sample example (human annotation "FAKE").

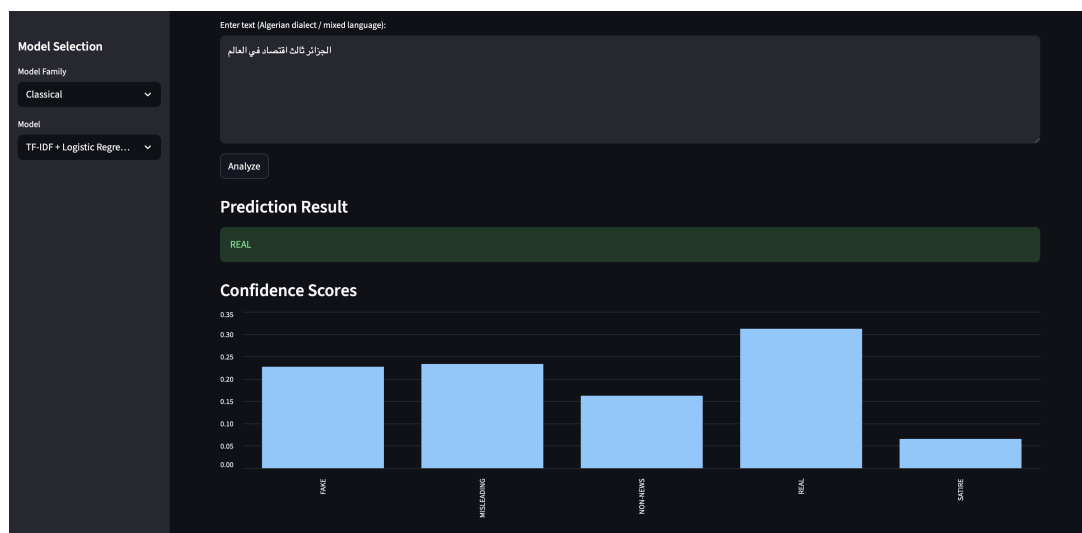


Figure 20: Screenshot demonstrating the result of our system on a sample example (human annotation "REAL").

Appendix B: Detailed Corpus Statistics

Table 9: Detailed Statistical Profile of the corpus

Metric Group / Feature	Value / Statistic
<i>Core Dimensions</i>	
Total Instances	22,391
Total Tokens	520,212
Vocabulary Size ($ V $)	113,105
Type-Token Ratio (TTR)	0.2174
<i>Text Length (Words)</i>	
Average Sentence Length	23.23
Median Sentence Length	14.00
Standard Deviation	33.86
<i>Linguistic Markers (Average per Text)</i>	
Punctuation Density	0.0594
Emoji Frequency	0.27
Hashtag Frequency	0.16
Letter Ratio	0.7914
<i>Annotation and Quality</i>	
Number of Annotators	6
Single Annotator Decision Rate	89.0%
Unanimous Agreement Rate	4.5%
Majority Agreement Rate	2.5%

Appendix C: Individual Contributions

This appendix details the specific contributions of each team member throughout the project lifecycle, encompassing data collection, model development, and manuscript preparation.

Manel AIT-SAID

Data Collection & Processing:

- Iter-Phase 1-1: Initial Data Collection
- Iter-Phase 1-2: Data Labeling of Phase 1-1
- Iter-Phase 2-1_2-2: Data Collection & Labeling
- Iteration 3: Data Augmentation
- Data Leakage Prevention Strategy + Execution

Paper Contributions:

- System Design
- Appendix A

Mohammed Abdelkhalek BENKINIOUAR

Data Collection & Processing:

- Iter-Phase 1-1: Initial Data Collection
- Iter-Phase 1-2: Data Labeling of Phase 1-1
- Iter-Phase 2-1_2-2: Data Collection & Labeling
- Iteration 3: Data Augmentation

Model Development:

- Model 1.1: TF-IDF + Logistic Regression
- Model 1.2: TF-IDF + Support Vector Machine

Paper Contributions:

- Related Work

Omar Ziyad CHAALEL

Data Collection & Processing:

- Iter-Phase 1-1: Initial Data Collection
- Iter-Phase 1-2: Data Labeling of Phase 1-1
- Iter-Phase 2-1_2-2: Data Collection & Labeling
- Iteration 3: Data Augmentation
- Data Cleaning Strategy + Execution

Paper Contributions:

- Abstract
- Introduction
- Conclusion
- Review and adjustment of paper content.

Rahil GHANEM

Data Collection & Processing:

- Iter-Phase 1-1: Initial Data Collection
- Iter-Phase 1-2: Data Labeling of Phase 1-1
- Iter-Phase 2-1_2-2: Data Collection & Labeling
- Iteration 3: Data Augmentation

Model Development:

- Model 4.1: mBERT Fine-tuned
- Model 4.2: XLM-RoBERTa-large + Data Augmentation

Paper Contributions:

- Dataset Building

Houda RAHMANI**Data Collection & Processing:**

- Iter-Phase 1-1: Initial Data Collection
- Iter-Phase 1-2: Data Labeling of Phase 1-1
- Iter-Phase 2-1_2-2: Data Collection & Labeling
- Iteration 3: Data Augmentation

Model Development:

- Model 2.1: Convolutional Neural Network (CNN)
- Model 2.2: Bidirectional LSTM

Implementation:

- GUI Implementation

Paper Contributions:

- Results and Analysis
- Discussion

Safia TIFOUR**Data Collection & Processing:**

- Iter-Phase 1-1: Initial Data Collection
- Iter-Phase 1-2: Data Labeling of Phase 1-1
- Iter-Phase 2-1_2-2: Data Collection & Labeling
- Iteration 3: Data Augmentation

Model Development:

- Model 3.1: DziriBERT Fine-tuned
- Model 3.2: XLM-RoBERTa-base Fine-tuned

Paper Contributions:

- Results and Analysis
- Discussion

Note: All team members participated in collaborative discussions, project planning meetings, and peer review of the manuscript. The contributions listed above represent the primary responsibilities undertaken by each member.